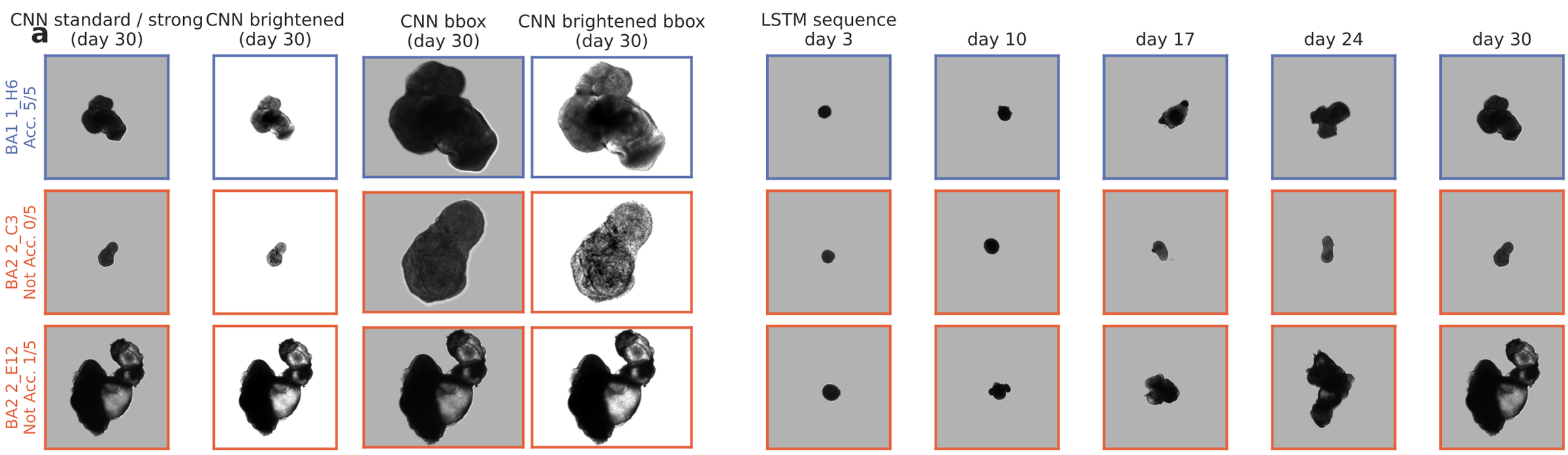
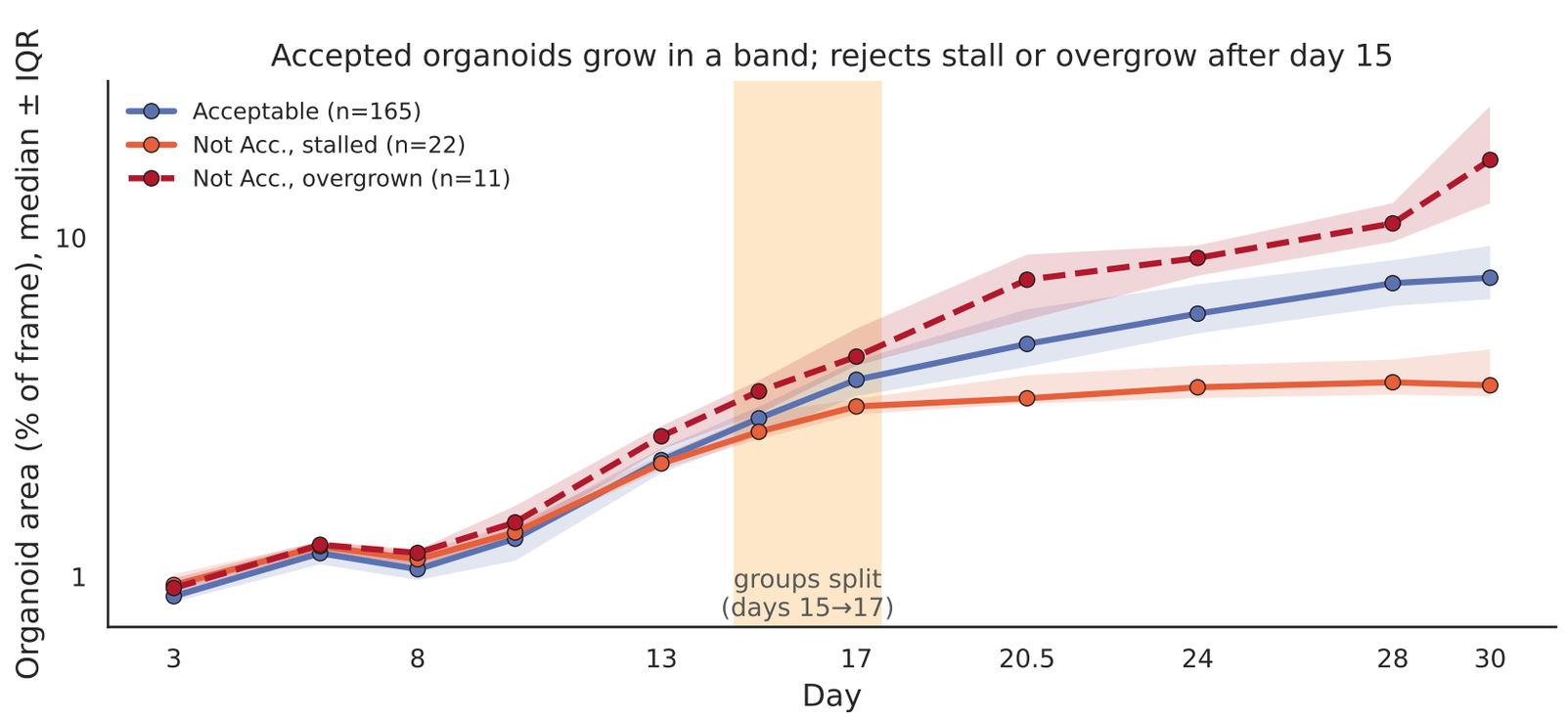
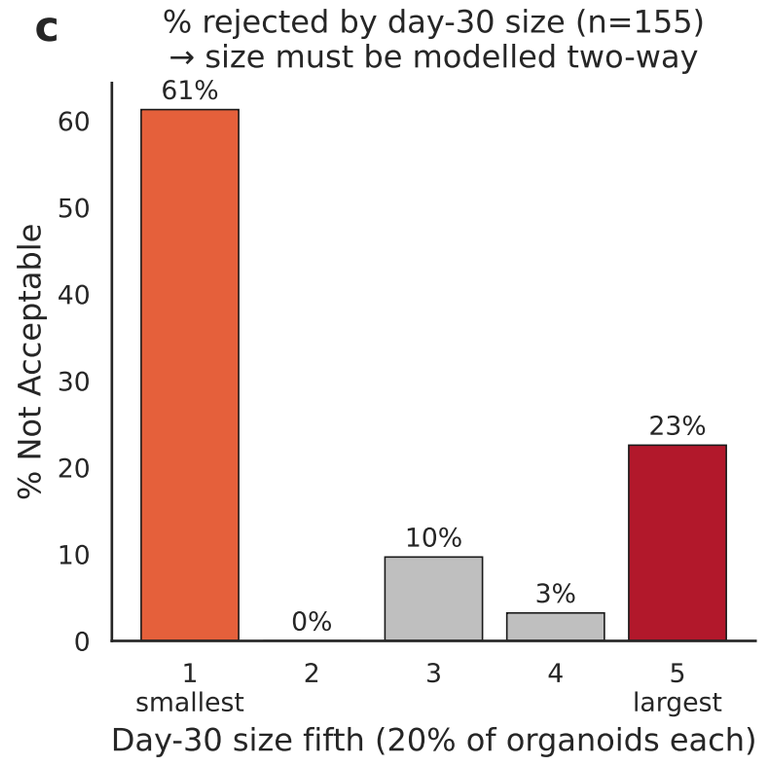
a


CNN standard / strong (day 30)
CNN brightened (day 30)
CNN bbox (day 30)
CNN brightened bbox (day 30)
LSTM sequence day 3
day 10
day 17
day 24
day 30
BA1 1_H6 Acc. 5/5
BA2 2_C3 Not Acc. 0/5
BA2 2_E12 Not Acc. 1/5

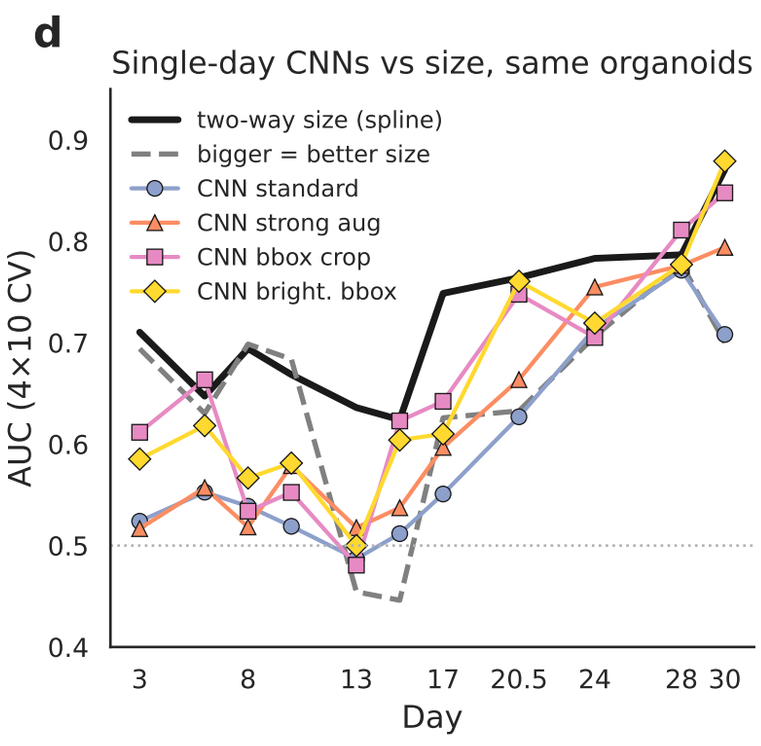
b


Accepted organoids grow in a band; rejects stall or overgrow after day 15
Acceptable (n=165)
Not Acc., stalled (n=22)
Not Acc., overgrown (n=11)
Organoid area (% of frame), median ± IQR
groups split (days 15→17)
Day

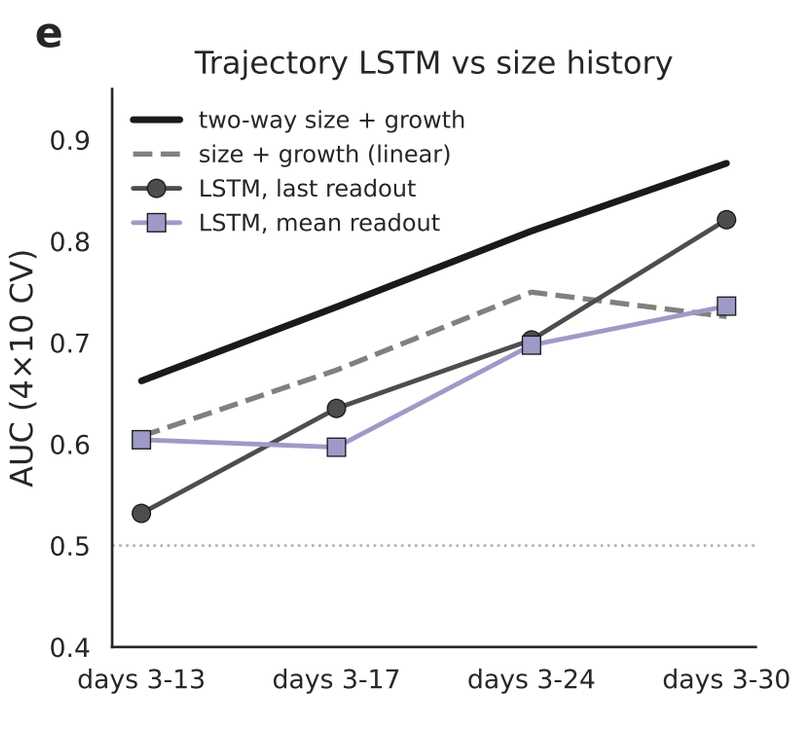
c


% rejected by day-30 size (n=155)
→ size must be modelled two-way
% Not Acceptable
61%
0%
10%
3%
23%
1 smallest
5 largest
Day-30 size fifth (20% of organoids each)

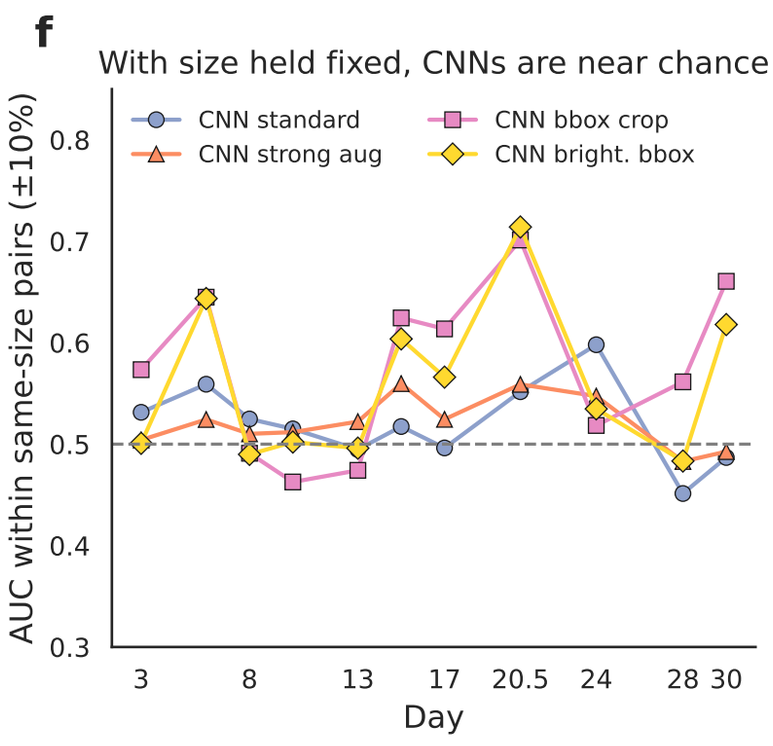
d


Single-day CNNs vs size, same organoids
two-way size (spline)
bigger = better size
CNN standard
CNN strong aug
CNN bbox crop
CNN bright. bbox
AUC (4×10 CV)
Day

e


Trajectory LSTM vs size history
two-way size + growth
size + growth (linear)
LSTM, last readout
LSTM, mean readout
AUC (4×10 CV)
days 3-13
days 3-17
days 3-24
days 3-30

f


With size held fixed, CNNs are near chance
CNN standard
CNN strong aug
CNN bbox crop
CNN bright. bbox
AUC within same-size pairs (±10%)
Day

g

Gain over two-way size + plate
CNN std
CNN strong
CNN bbox
CNN br. bbox
LSTM last
LSTM mean
Δ AUC
Day (LSTM: last day of window)

h

Bright. bbox CNN, day 30: small or big = bad
Acceptable
Not Acceptable
CNN P(Acceptable), OOF mean
Day-30 organoid area (×1000 px)